

Foundations of Machine Learning (AI2000/AI5000): Assignment-4
Released on: 10th November, 2025
Deadline: 20th November, 2025, EoD (11:59PM, IST)
Maximum Marks: 25M

A. Instructions

- Answer all questions. We encourage best coding practices by not penalizing (i.e. you may not get full marks if you make it difficult for us to understand). Hence, use intuitive names for the variables, and comment your code wherever necessary. Use the markdown cells in the notebook for briefly explaining the objective of a code cell or reporting the observations if asked in the question.
- It is expected that you work on these problems individually. If you have any doubts please contact the TA or the instructor no later than 2 days prior to the deadline.
- Use of LLMs are strictly prohibited. If found, you will be heavily penalized in the respective questions.
- **Implement all algorithms in this assignment from scratch using first principles.** You may only use basic mathematical functions from libraries such as `numpy` (e.g., `np.mean()`, `np.std()`, `np.sum()`, `np.linalg.eig()`, `np.sqrt`, `np.log`, and basic array operations). The use of `matplotlib` for visualization and `numpy` for fundamental numerical operations. The goal is to demonstrate a fundamental understanding of the algorithms, not their library-based application.
- **Students are required to submit a Python notebook titled A4.<your-roll-number>.ipynb, together with a PDF document containing all relevant mathematical derivations as specified in the assignment.**

B. Problems

1. Dimensionality Reduction (15 Marks)

In this question, you will study and derive the **Principal Component Analysis (PCA)** formulation and apply it conceptually to face recognition using the given face dataset.

Background

The dataset contains N face images. Each original image I_k is a gray-scale image of size $h \times w$ pixels, where each pixel has a single intensity value.

(a) Data Representation (2M)

In this section, you will work with the given face dataset to understand how images are represented mathematically and organized into a data matrix for computational analysis.

Your Tasks:

1. **Convert Images to Vectors:** For each image I_k in the dataset, you must convert the 2D pixel array into a 1D vector $\mathbf{x}_k$ in $\mathbb{R}^{hw}$. This process is called **vectorization**. You should achieve this by stacking the columns of the image matrix from left to right.

Formally, for an image matrix:

$$I_k = \begin{bmatrix} I_k(1,1) & I_k(1,2) & \cdots & I_k(1,w) \\ I_k(2,1) & I_k(2,2) & \cdots & I_k(2,w) \\ \vdots & \vdots & \ddots & \vdots \\ I_k(h,1) & I_k(h,2) & \cdots & I_k(h,w) \end{bmatrix}$$

The corresponding vector is:

$$\mathbf{x}_k = \begin{bmatrix} I_k(1, 1) \\ I_k(2, 1) \\ \vdots \\ I_k(h, 1) \\ I_k(1, 2) \\ \vdots \\ I_k(h, w) \end{bmatrix} \in \mathbb{R}^{hw}$$

where $I_k(i, j)$ is the gray-scale intensity at row i and column j .

2. **Construct the Data Matrix:** Once all images are vectorized, you must define the data matrix X . This matrix should be constructed by placing each vector $\mathbf{x}_k$ as a column. The final data matrix X will have the following form:

$$X = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_N \\ | & | & \cdots & | \end{bmatrix} \in \mathbb{R}^{hw \times N}$$

Each column of X therefore represents one complete face image from the dataset.

(b) Mean Centering and Covariance Matrix (3M)

In this section, you will formalize the key mathematical objects needed for PCA.

- (a) **Mean Face:** $\mu = \frac{1}{N} \sum_{i=1}^N I_i \in \mathbb{R}^{hw}$
- (c) **Covariance matrix:** $C = \frac{1}{N} XX^T \in \mathbb{R}^{hw \times hw}$
- (b) **Mean-centered images:** $x_i = I_i - \mu \in \mathbb{R}^{hw}$
for $i = 1, \dots, N$

(c) PCA Eigenvalue Problem (4M)

In this section, you will formalize the core mathematical concepts of PCA as applied to face recognition.

- (a) **Eigenvalue Problem:** **State** the eigenvalue problem for the covariance matrix, defining the eigenvectors and eigenvalues, and **implement** its computation to obtain the principal components.
- (b) **Variance Interpretation:** Explain why eigenvectors with the largest eigenvalues capture the directions of maximum variance in the data.
- (c) **Dimensionality Reduction:** Briefly explain how selecting the top- K eigenvectors enables dimensionality reduction while preserving the most important data features.

(d) Projection and Reconstruction in PCA Subspace (6M)

Let $U_k = [u_1, u_2, \dots, u_k] \in \mathbb{R}^{hw \times k}$ denote the matrix formed by the top- k eigenvectors.

- (a) **Projection and Reconstruction:**

- **Formulate** the projection operation mathematically and provide its computational **implementation** to map a test face image onto the principal component subspace, resulting in a lower-dimensional feature vector.
- **Formulate** and **implement** the reverse transformation that maps a feature vector from the principal component subspace back to the original image space, producing a reconstructed face image.
- Evaluate the quality of the reconstruction by calculating the **reconstruction error** between each original test image and its reconstructed approximation.

(b) **Experimental Analysis:**

- Experiment with different values of k (e.g., 10, 50, 100, 200) on test data
 - Plot reconstruction error vs. number of components k
 - Visualize original and reconstructed faces for different k values
 - Write your conclusions and inferences about the trade-off between compression and reconstruction quality
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2. Unsupervised Learning (10M)

In this question, you will implement and compare different clustering algorithms on 2D datasets to understand their properties, limitations, and appropriate applications.

Download the following three 2D datasets provided for this assignment:

- **Dataset A:** `dataset_A.csv`
- **Dataset B:** `dataset_B.csv`
- **Dataset C:** `dataset_C.csv`

Each dataset contains points with (x, y) coordinates in CSV format.

Part (a): K-Means Clustering (5M)

(a) **Theoretical Analysis:**

- Derive the objective function that K-Means minimizes and explain the optimization process
- Discuss the time and space complexity of your implementation
- Explain the limitations of K-Means regarding cluster shapes and sizes

(b) **Experimental Analysis:**

- Apply K-Means to all three datasets with appropriate K values
- Visualize clustering results with cluster boundaries and centroids
- Evaluate clustering quality using validation metrics (Within-Cluster Sum of Squares)

Part (b): Gaussian Mixture Models (GMM) (5M)

(a) **Theoretical Analysis:**

- Derive the complete data log-likelihood function for GMM
- Mathematically formulate the E-step and M-step
- Explain the concept of soft assignments and how they differ from hard assignments in K-Means

(b) **Experimental Analysis:**

- Apply GMM to all three datasets with appropriate numbers of components
 - Visualize results with probability contours and cluster assignments
 - Analyze the convergence of the log-likelihood during EM iterations
 - Compare the performance and results of GMM with K-Means on each dataset
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